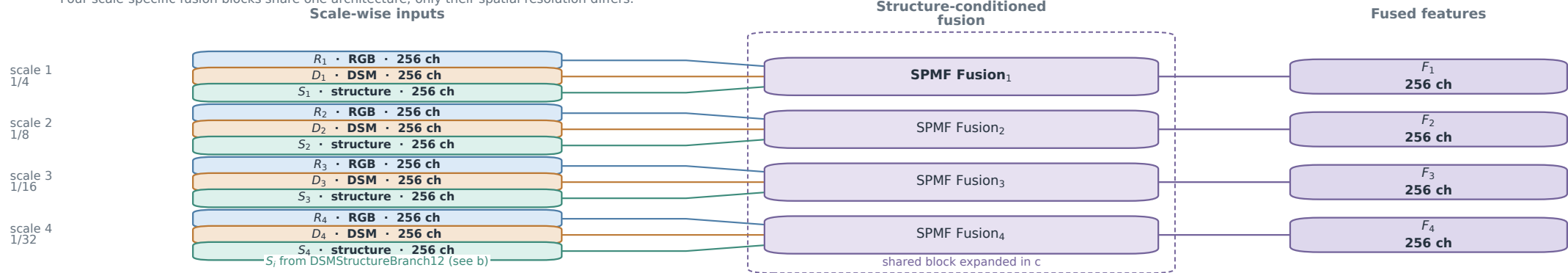


SPMF21: multi-scale structure-conditioned multimodal fusion

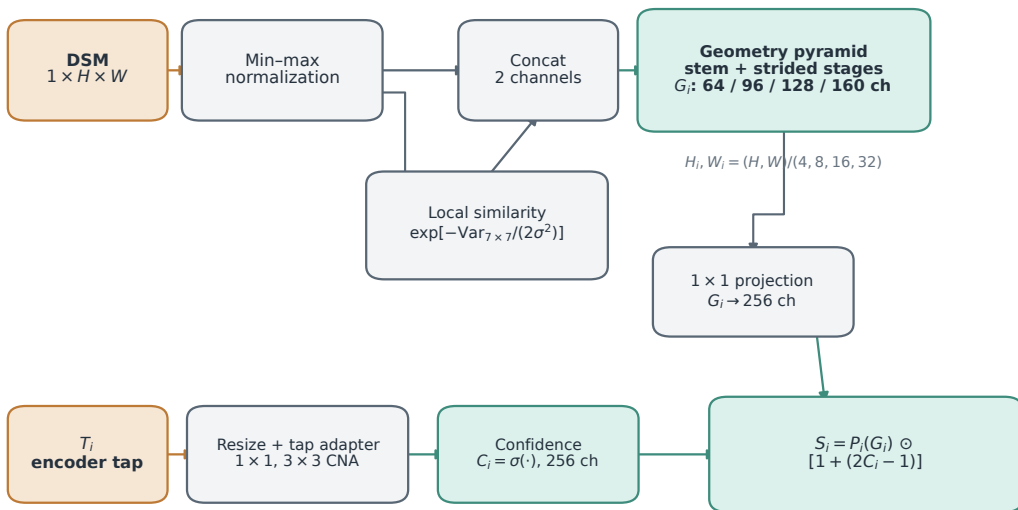
a SPMF21 overview

Four scale-specific fusion blocks share one architecture; only their spatial resolution differs.



b DSMStructureBranch12

DSM geometry and a scale-matched encoder tap jointly form each structure prior.



CNA: convolution + normalization + GELU • tap gradients are detached by default

c Repeated single-scale SPMF fusion block

A shared block computes 64-channel evidence and routes the original modality features.

$$z' = z(1 + \gamma) + \beta$$

